Learning path development for plant data management

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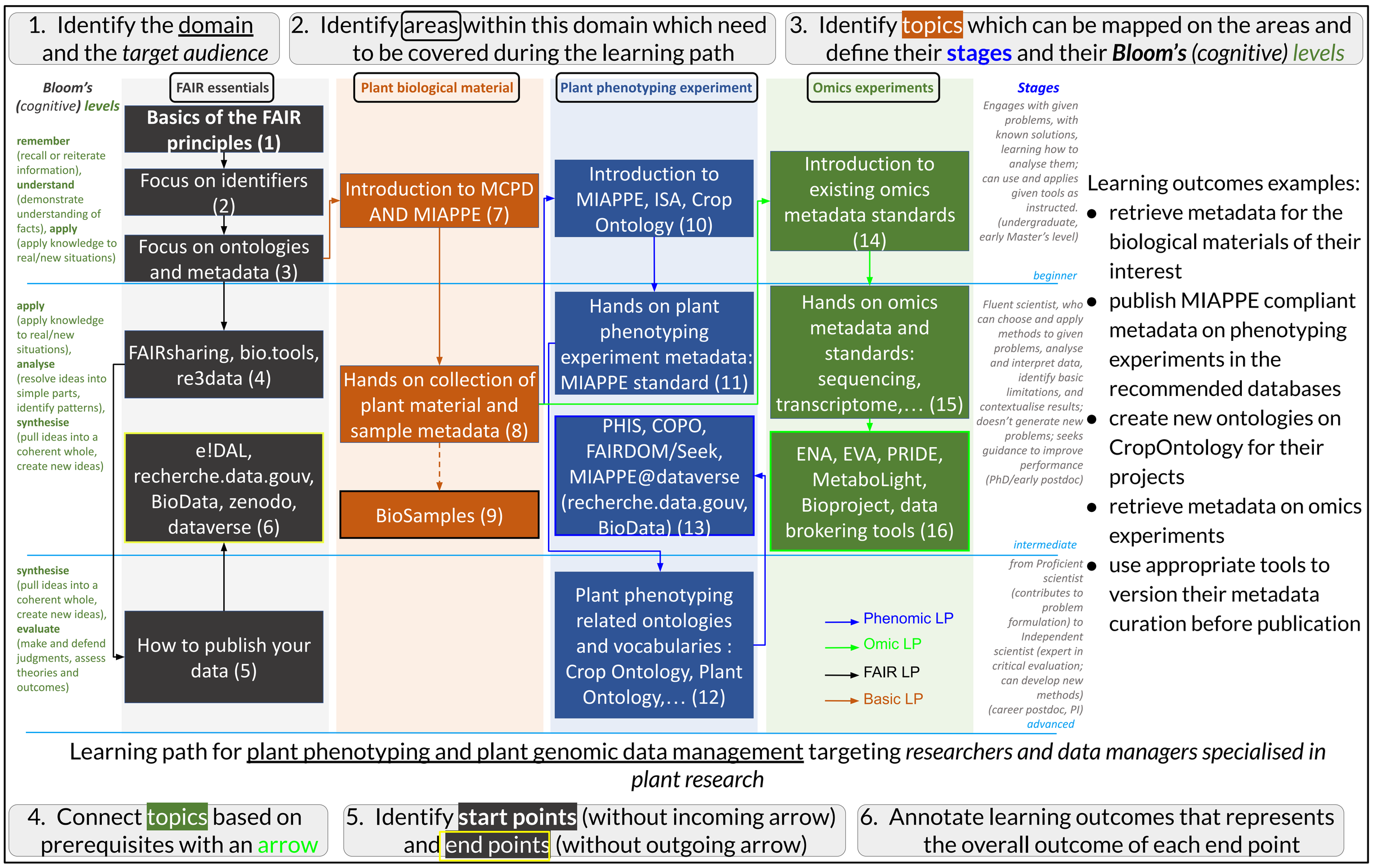
A learning path was developed for plant phenotyping and plant genomic data management in the frame of the ELIXIR CONVERGE project as a way to analyse the training gaps in this area and to the improve training resources of the plant Science Community.

A learning path is a collection of courses that need to be followed in a specific order to acquire knowledge and skills on a precise domain. This work has been achieved during the 2022 ELIXIR BioHackathon, in which the ELIXIR training platform was testing a template that can be used to develop new learning paths.

Existing training material found in TeSS (Training eSupport System) or gathered by the ELIXIR plant community was then mapped on the learning path in order to identify gaps to be addressed.
The FAIRness of each training material gathered has been assessed following criteria derived from Garcia et al. (2020) Ten simple rules for making training materials FAIR https://doi.org/10.1371/journal.pcbi.1007854. As a conclusion a FAIR score has been assigned to each training material based on these criteria



Training	No. of training found	FAIR score	Training	No. of training found	FAIR score	Training	No. of training found	FAIR score	Training	No. of training found	FAIR score
(1)	4	[12-14]/17	(5)	2	[7.5-15]/17	(9)	1	6.5/17	(13)	1	6.5/17
(2)	3	[8-11]/17	(6)	1	12/17	(10)	1	5.5/17	(14)	0	
(3)	2	13/17	(7)	3	[5-12]/17	(11)	1	8/17	(15)	0	
(4)	2	[4-13]/17	(8)	0		(12)	1	7/17	(16)	6	[4.5-8]/17

Mapping of training materials on the learning path. Numbers in brackets represent topics on the learning path above with each colour covering a precise domain. The FAIR score is assessed on 17 criteria and intervals are given when multiple training materials are mapped to the topic (from lowest to highest score). This mapping highlights the topics which are not covered as much as the others in terms of training material (e.g. topic (1) is mapped to 4 training materials and topic (8) is not covered by any training material) but also the fluctuation between topics in terms of FAIR score (topics on the FAIR essentials domain have higher FAIR scores than other domain's topic).

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